


**Fig. 1.12** The Gay-Berne potential depends strongly on the relative orientations of the rigid bodies, after a figure in [143]

in providing a precise parameterization [187]: using a general model for water (one not optimized for their specific experiment) the melting point of ice was underestimated by approximately 20 °K.

## 1.2 The N-Body Problem

In this book, we shall frequently use a compact, vectorial notation, where $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$ represent vectors of the positions and velocities, and $\boldsymbol{M}$ is a diagonal mass matrix, so the equations of motion (1.2) become

$$\boldsymbol{M}\frac{d^2}{dt^2}\boldsymbol{q} = \boldsymbol{F}(\boldsymbol{q}) = -\nabla U(\boldsymbol{q}). \tag{1.3}$$

If the N particles are confined to the real line, then $\boldsymbol{q} = (q_1, q_2, \ldots, q_{\mathrm{N}})$ and the components of the force $\boldsymbol{F}$ are $F_i = -\partial U/\partial q_i$. For motion in $\mathbb{R}^3$, we write $\boldsymbol{q}_i = (q_{i,x}, q_{i,y}, q_{i,z})$ for the position vector of the $i$th particle, and the force acting on this particle is $\boldsymbol{F}_i = -\nabla_{\boldsymbol{q}_i} U = -\partial U/\partial \boldsymbol{q}_i$, that is, the vector with components $F_{i,x} = -\partial U/\partial q_{i,x}, F_{i,y} = -\partial U/\partial q_{i,y}, F_{i,z} = -\partial U/\partial q_{i,z}$. We may also write $x_k, y_k, z_k$ for the $x$, $y$ and $z$ components of the position of particle $k$. We let $\mathrm{N_c}$ represent the total number of position coordinates of our system, thus for a system of N atoms in three space dimensions there are $\mathrm{N_c} = 3\mathrm{N}$ coordinates. In the typical case of 3-dimensional space, we view $\boldsymbol{q}$ as a $\mathrm{N_c} = 3\mathrm{N}$-dimensional vector and write $\boldsymbol{M} = \mathrm{diag}(m_1, m_1, m_1, m_2, m_2, m_2, \ldots, m_{\mathrm{N}}, m_{\mathrm{N}}, m_{\mathrm{N}})$.

The number of local directions in which the configurational (position) state can be varied is called the number of *degrees of freedom* $\mathrm{N_d}$. For the N-body system in $\mathbb{R}^3$ without additional constraints there are $\mathrm{N_d} = \mathrm{N_c} = 3\mathrm{N}$ degrees of freedom. If $r$ independent constraints are present the number of degrees of freedom is $\mathrm{N_d} = \mathrm{N_c} - r$.

The total energy of the N-body system is a function of positions and velocities,

$$E(\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_{\mathrm{N}}, \dot{\boldsymbol{q}}_1, \dot{\boldsymbol{q}}_2, \ldots, \dot{\boldsymbol{q}}_{\mathrm{N}}) = \sum_{j=1}^{\mathrm{N}} \frac{m_j \|\dot{\boldsymbol{q}}_j\|^2}{2} + U(\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_{\mathrm{N}}). \tag{1.4}$$